

# Group Production and Operations Reporting Standard

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| <b>Owner</b>          | Group Operational Excellence |
| <b>Classification</b> | Internal                     |

## 1. Purpose

This standard defines how operational throughput, uptime, energy and emissions are measured, recorded, and reported within the Group. It is the authoritative reference for the operations\_daily Delta table maintained in Unity Catalog.

## 2. Reporting Units

Throughput shall be reported in the standard unit appropriate to the site type:

- Offshore Platforms and Onshore Fields: thousand barrels per day (kbbbl/d) for liquids, million standard cubic feet per day (MMscf/d) for gas.
- Refineries and Trading Desks: thousand barrels per day (kbbbl/d).
- LNG Plants: million tonnes per annum (MTPA).
- Petrochemical Plants: kilotonnes per day (kt/d).
- Retail Clusters: kilolitres per day (kL/d).
- Marine Terminals: thousand barrels per day (kbbbl/d).

## 3. Data Submission

Site Operations is responsible for submitting daily data by 09:00 local time the following day. Submissions land in the bronze schema, are validated by Group Data Engineering, and promoted to silver.

The gold table operations\_daily presents the assured Group-wide view used by leadership dashboards, the Genie Space and the AI Co-Pilot.

## 4. Data Quality Rules

Each row in operations\_daily shall contain a non-null operation\_date, site\_id, business\_unit, throughput\_value, throughput\_unit, uptime\_pct, energy\_consumed\_mwh, and emissions\_tco2e.

throughput\_value must be  $\geq 0$ . Negative values are rejected at the bronze layer.

uptime\_pct must be in  $[0, 100]$ . Values outside the range are flagged for review.

emissions\_tco2e shall be calculated using the latest approved emission factor for the site\_type.

## 5. Variance Investigation

Variances exceeding 10% versus throughput\_target shall include a written explanation in the variance\_notes column at the silver layer. Variances exceeding 15% trigger an automatic notification to the BU Operations Manager.

## **6. Auditability and Lineage**

All transformations from bronze to silver to gold shall be authored as Lakeflow Spark Declarative Pipelines and registered in Unity Catalog. Column-level lineage shall be available for every gold column.

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